# KCES'S
# College of Engineering and Management , Jalgaon

## Software Build (Mini Model) Competition

# TECHNO-STORM 2K26

**Project Name**

Vaultora

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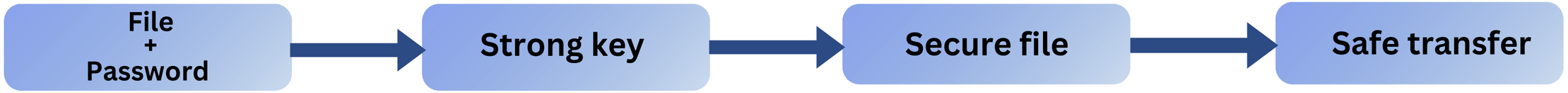
# PROBLEM STATEMENT

- Data breaches are rising with  most files (images, PDFs, videos) travel unencrypted.
- No single lightweight tool encrypts all file types.
- Simple password hashing is broken need modern key derivation.

# SOLUTION

→ **Vaultora** ( web-based ) encryption/decryption system for any file type.




→ **Works in Web Browser** — No installation needed for end users.

| Layer | Technology | Why |
|---|---|---|
| Frontend | HTML, CSS, JS | Simple, universal UI |
| Backend | Python + Flask | Lightweight REST API |
| Encryption | Fernet (cryptography) | AES-128-CBC + HMAC |
| Key Derivation | PBKDF2HMAC + SHA256 | Brute-force resistant |
| Encoding | Base64 | Binary-safe transport |

# FILE BASE

- TEXT
- IMAGE
- VIDEO
- PDF

## Encryption

File + Password
↓
Salt (os.urandom 16 bytes)
↓
PBKDF2HMAC → 32-byte key → base64url encode
↓
Fernet(key).encrypt(file_bytes)
↓
Encrypted token → Base64 encode → Send/Save

## Decryption

Encrypted file + Password + Salt
↓
Re-derive same key via PBKDF2HMAC
↓
Fernet(key).decrypt(token)
↓
Original file bytes → restored file

**Browser — HTML/CSS/JS**

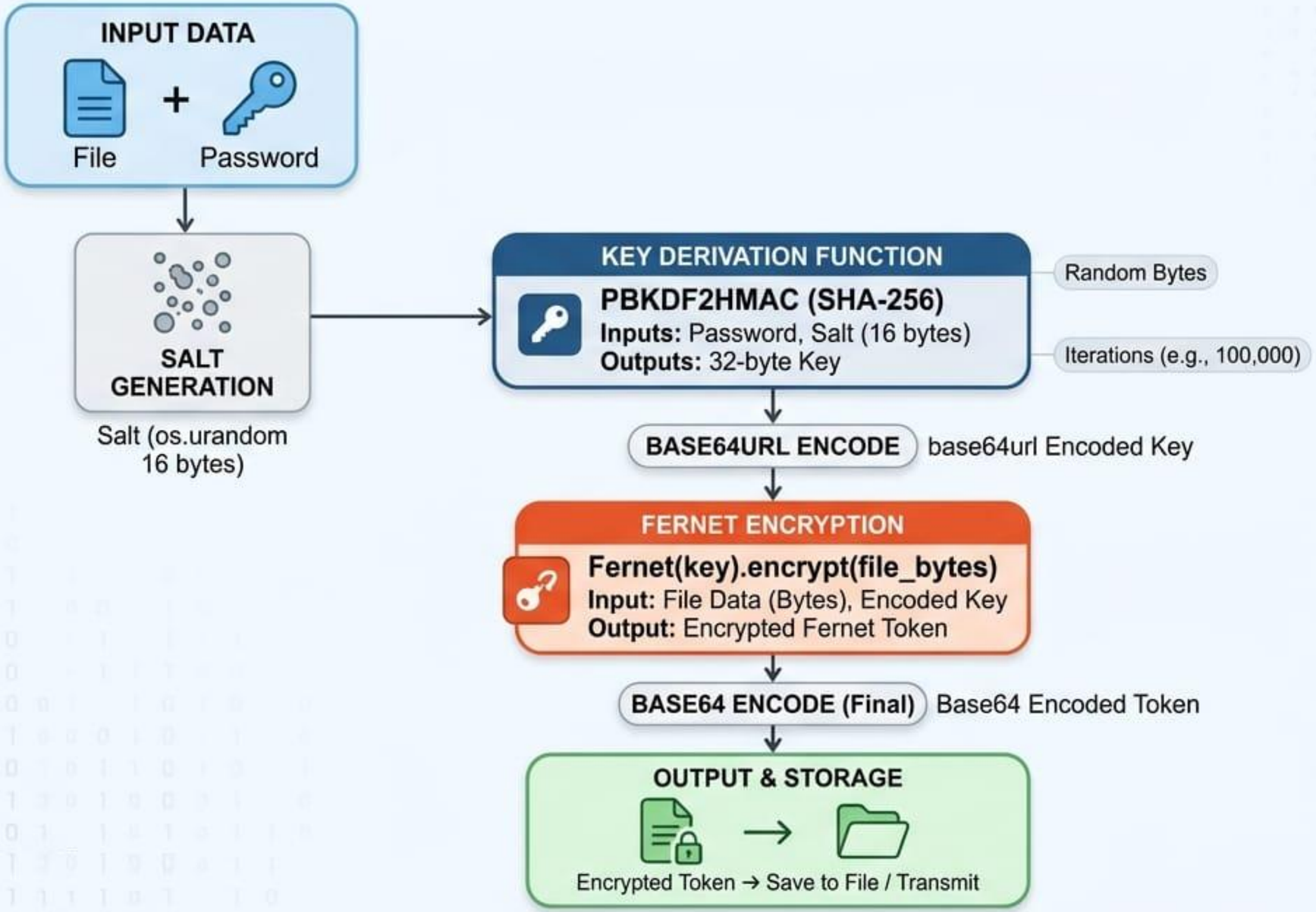
↓ HTTP (multipart form + JSON)

**Flask Routes — /encrypt /decrypt**

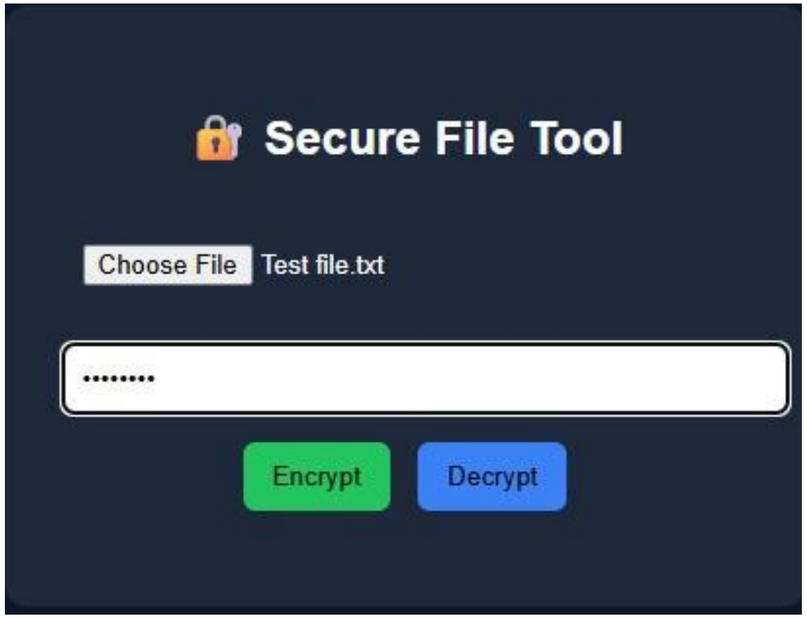
↓ Python function calls

**Crypto Engine — PBKDF2HMAC + Fernet + Base64**

# SECURE FILE ENCRYPTION WORKFLOW (PBKDF2-HMAC-SHA256 & FERNET)



Application



Wrong Password
during Decryption

## 🔒 ENCRYPT → 🔑 DECRYPT

### Always Free

- ✅ All file types
- ✅ Login only for Subscription
- ✅ Instant download
- ✅ Unlimited encryptions

### Choose your plan

₹1 per file
₹299 per month
₹999 per year
✅ Unlimited decrypt

## *Industries That Need Us Most:*

- SecureVault – Smart Data Protection Platform
- Legal — LexVault
- Fintech — DocuSafe
- Personal Privacy — SafeSend
- PixelLock - Freelancers

# Reference Links & Comparison

## Reference Websites :

- futuremarketinsights.com
- mordorintelligence.com
- ibm.com/security/data-breach
- saasworthy.com
- comparecheapssl.com
- grandviewresearch.com
- cloudwards.net
- github.com/Encryption-Decryption-Tool/

| Feature | Your App | AxCrypt | Tresorit | Boxcryptor | Cryptomator |
|---|---|---|---|---|---|
| Encryption Algorithm | Fernet (AES-128 + HMAC) | AES-128 / AES-256 | AES-256 | AES-256 | AES-256 |
| Key Derivation | PBKDF2HMAC (480K iter) | Proprietary | Proprietary | Proprietary | SCrypt |
| Text File Support | ✅ | ✅ | ✅ | ✅ | ✅ |
| Image Encryption | ✅ | ✅ | ✅ | ✅ | ✅ |
| PDF Encryption | ✅ | ❌ | ✅ | ✅ | ✅ |
| Video Encryption | ✅ | ❌ | ❌ | ❌ | ✅ |
| Web-Based (No Install) | ✅ | ❌ | ❌ | ❌ | ❌ |